

Varshini Akula

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SUMMARY

An exceptional software engineer with proven technical experience from exposure at a leading global company, adept at solving complex business challenges through innovative software engineering and AI-driven solutions.

EDUCATION

Arizona State University

Master of Science in Computer Science

Tempe, AZ

Aug 2025 - May 2027

Jawaharlal Nehru Technological University

Bachelor of Technology in Computer Science and Engineering (Data Science) (9.49/10 CGPA)

Hyderabad, India

May 2024

PROFESSIONAL EXPERIENCE

JPMorgan Chase, Hyderabad, India: Software Development Engineer I

Jul 2024 – Aug 2025

- Designed and shipped a production full-stack conversational AI system using **Rasa NLU, Python services, AWS, and React** for document storage onboarding, automating metadata extraction and eliminating **320+ hours of manual data** entry per quarter.
- Built **Java, SQL, and AWS microservices** to automate document storage record CRUD operation across Consumer & Community Banking, **cutting storage provisioning time by 40%**. Engineered backend Java applications leveraging **MVC architecture and Spring Boot** to seamlessly integrate **microservices and REST APIs**, while designing optimized data views using **SQL** for scalable and efficient data access.
- Hardened production observability with **structured logging, distributed tracing, and health-check endpoints** feeding **real-time monitoring dashboards** for services under a **99% uptime SLA**.

JPMorgan Chase, Hyderabad, India: Software Engineering Program Intern

Jan 2024 – Jun 2024

- Engineered a full-stack LLM-powered incident chatbot with **Python, LLMs, and TypeScript** that summarized tickets, generated severity-based runbooks, and **reduced mean incident query resolution time by 60%**.
- Developed a real-time data streaming application with **Java, Spring Boot and Apache Kafka** for the Global Technology & Innovation team, lowering cross-system propagation latency for **70+ downstream applications**.

PROJECTS

Hybrid Multi-Retriever Orchestration for Long-Term Conversational Memory, Research Project

Fall 2025

- Built a **RAG orchestration** system in Python, routing queries across **BM25, TF-IDF, FAISS, SVM, and time-weighted** retrievers using user intent-based and rule-based decision logic for long memory context queries.
- Improved **Recall@5 from 0.65 to 0.83 (+18 pts)** on LongMemEval with **Qwen2.5-7B** and **+4 pts on Locomo** with **LLaMA-3.1-8B** through **few-shot prompt engineering**.

Construction Project Intelligence

Spring 2026

- Built a full-stack **FastAPI & React** document intelligence platform with **RBAC, RAG Q&A with citations, PostgreSQL-backed indexing**, RFI classification, and real-time analytics dashboards; **reduced manual lookup time by 95%**.

FlowMind: Visual Design and Debug Tool for AI Agents

Spring 2026

- Integrated Building a visual AI workflow platform for designing, debugging, and deploying AI assistants with **graph-based conversation flows, RAG citations, live analytics dashboards, and hosted chat deployment** using **Next.js 15, React 19, and TypeScript**.

TECHNICAL SKILLS

Programming Languages: Python, Java, TypeScript, JavaScript, HTML, CSS, SQL, C, Bash

Tools/Frameworks: FastAPI, Maven, Git, GitHub, Bitbucket, JIRA, Jenkins, Postman, Swagger, Docker, Kubernetes, Spring, Spring Boot, REST, Apache Kafka, WebSocket, JUnit, Mockito, Selenium, Next.js, Tailwind CSS, Linux, NVIDIA A100 GPU clusters

AI/ML: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LangChain, OpenAI APIs, Rasa NLU, RAG Systems, LLM Fine-Tuning

Databases: SQL, NoSQL, MySQL, Oracle DB

Core Competencies: Software Requirements and Specifications, Cloud Computing, Object Oriented Programming, Data Structures and Algorithms, Database Management Systems

Certifications: AWS Certified Cloud Practitioner (Dec 2024)

PUBLICATIONS

Evaluating the Performance of Fake Face Detection Methods in Forensics, IJITCE, Jun 2024

Optimizing Prediction of Security Data Using Feature Selection and Ensembling, IJITCE, May 2023